Giuditta Pezzotta · Paolo Gaiardelli ·
Chiara Cimini · Roberto Sala · David Romero ·
Jannicke Baalsrud-Hauge
Editors

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*Editors*
Giuditta Pezzotta
University of Bergamo
Bergamo, Italy

Paolo Gaiardelli
University of Bergamo
Bergamo, Italy

Chiara Cimini
University of Bergamo
Bergamo, Italy

Roberto Sala
University of Bergamo
Bergamo, Italy

David Romero
Tecnológico de Monterrey
Mexico City, Mexico

Jannicke Baalsrud-Hauge
KTH Royal Institute of Technology
Stockholm, Sweden

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# Wear- and Time-Aware Pricing with Equity Guardrails for IoT-Tracked Shared Tools

Waqas Ahmed[1,2,3(✉)], Hans-Åke Sundberg[1], and Robert Shorten[2]

[1] Husqvarna Group, Huskvarna, Sweden
waqas.ahmed@ju.se
[2] Dyson School of Design Engineering, Imperial College London, London, United Kingdom
[3] School of Engineering, Jönköping University, Jönköping, Sweden

**Abstract.** Neighborhood tool sharing requires fair and predictable governance. This work presents an interpretable, component-wise pricing and coordination framework that integrates time- and wear-aware charges, bounded behavior multipliers, queue compensation, GPS compliance, and predictability caps (spare-part, replacement, and cumulative equity guardrails). Instantiated on the Husqvarna 540iXP and evaluated via Monte Carlo simulation with 200 households over 60 days across capacities $M \in \{5,10,20,40\}$ and varying adversarial fractions, the framework is benchmarked against time-only and ablated baselines. Results indicate that capacity is a primary lever: scarce supply yields near-saturation utilization and heavy queues, while higher capacity can improve served-household fairness (with transitional non-monotonicity in intermediate regimes). At $M = 40$, queues are effectively cleared (mean $\approx$0.01d; median $\approx$0; p95 $\approx$0.07d), with clipped utilization of $\approx$0.72 at $f_{\text{adv}} = 0$ and higher under adversarial activity. Mechanism richness can reduce fairness under tight capacity unless parameters are carefully calibrated; the equity guardrail truncates extreme exposure tails, improving predictability, especially under stressed regimes. Representative billed costs cluster around $\approx$75 SEK per lease ($95^{\text{th}}$ $\approx$275). Overall, the framework balances accountability with bounded household exposure and offers actionable levers for municipal and cooperative deployments.

**Keywords:** Distributed computing · smart systems · IoT · cyber-physical system

## 1 Introduction

Distributed resource allocation in smart communities is a central challenge in IoT systems [1]. Neighborhood tool-sharing programs illustrate this challenge vividly: households function as distributed nodes competing for a limited pool of shared tools, each with distinct usage patterns, reliability levels, and behavioral histories. Prior research on community resource sharing shows that decentralized peer-to-peer systems face similar coordination and contention challenges [2], while studies on neighborhood governance emphasize the need for explicit accountability mechanisms in collaborative settings [3]. Community-owned resource-sharing platforms likewise demonstrate that effective scheduling and access control require more than informal lending models [4].

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Although sustainability concerns motivate shared-resource systems [1, 5], tool sharing remains difficult to operationalize at scale. The circular and sharing economies highlight potential benefits through reduced underutilization and extended product lifetimes [6–8], and the Ellen MacArthur Foundation notes that many household tools are rarely used despite their material footprint [9].

Tool sharing is, at its core, a distributed micro-infrastructure problem: a shared resource must circulate efficiently among many autonomous nodes, delays must be managed, and misuse must be detected and discouraged. However, existing community tool-sharing initiatives frequently struggle with adoption. Studies identify several recurring barriers, households worry about tool condition, accountability for damage, and reliability of returns [10, 11], ownership is associated with autonomy, convenience, and status [12, 13], scheduling conflicts and uncertainty about availability reduce willingness to participate [14, 15], tool libraries struggle with underutilization, damage disputes, and inconsistent governance [16], limited visibility and digital-literacy barriers hinder adoption [17] and borrowing fees, deposits, and unpredictable penalties reduce trust in shared systems [15, 16, 18].

Therefore, tool sharing continues to face persistent adoption challenges, suggesting the need for an operational model that unifies sensing, incentives, and coordination into a transparent rule-set. The primary contribution of this work is methodological: an interpretable cyber-physical governance and pricing framework (formal cost model with caps/guardrails) and a simulation-based evaluation that tests its behavior under varying capacity and adversarial conditions. The core research question is:

*Under what supply regimes and behavioral conditions can an interpretable, sensor-informed, capped pricing-and-coordination mechanism bound household exposure while maintaining acceptable fairness and waiting-time performance?*

The sustainability relevance is indirect here, the framework is motivated by reduced underutilization and extended tool lifetime, but the evaluation focuses on service and economic outcomes (fairness, utilization, waiting time, and cost exposure) rather than direct environmental metrics.

## 2 Research Method

An inductive, problem-driven method rooted in industrial practice is adopted. Rather than theorizing in isolation, the study evaluates a practical governance framework for neighborhood tool sharing using (i) a battery-chainsaw case study, (ii) a formal cost model with interpretable components (time, usage, wear, inspection, delay, behavior, GPS, and caps), (iii) analytical queueing intuition to provide first-order capacity insights, and (iv) a Monte Carlo simulator that quantifies fairness, efficiency, and robustness under behavioral heterogeneity and adversarial activity.

A governance framework refers to the operational rule-set and mechanisms by which a neighborhood tool-sharing program coordinates access, computes charges, enforces compliance, and bounds household exposure using interpretable components. The framework was derived inductively by translating recurring adoption barriers into explicit design requirements and then mapping each requirement to a mechanism in the cost

model and evaluation metrics (detailed later). This problem-driven artifact construction and evaluation is consistent with design-science research framing and established inductive theory-building guidance [19, 20].

### 2.1 Chainsaw Case Study: 540iXP Battery Chainsaw

The Husqvarna 540iXP is used as a representative high-value shared tool. Typical tasks include pruning, felling trees, and preparing firewood. These operations expose the tool to both wear-inducing loads (runtime, vibration, and cutting engagement) and non-wear events (drops, impacts, and improper assembly). Wear-sensitive components include the motor, control unit, cutting chain, guide bar, and sprocket, while misuse-sensitive components include the chassis, oil tank, throttle lockout, and screws.

Figure 1 illustrates the tool and highlights wear-sensitive versus non-wear components. The chainsaw typifies household tools for which spare-part and replacement costs are significant and for which sensor telemetry can meaningfully inform usage and wear estimates.




**Fig. 1.** 540iXP battery chainsaw and component breakdown

### 2.2 System Architecture

Each chainsaw is instrumented with sensors measuring battery state of charge (SoC), runtime, oil-flow status, RPM/power, and GPS position (see Fig. 2). An on-device microcontroller on the 540iXP battery chainsaw streams telemetry to an edge node,

which performs lightweight feature extraction including normalized runtime, task features, load proxies, per-component remaining useful life (RUL) estimates, and associated confidence measures and forwards derived signals to the cloud coordination service.

The cloud service maintains global system state, including tool availability, queues, and behavior/compliance indicators, and executes an interpretable pricing mechanism comprising time-based charges, wear-based charges, delay penalties, behavior multipliers, queue compensation, and GPS-based compliance, subject to predefined caps. Per-lease subtotals and confidence metadata are logged to enable transparent auditing.

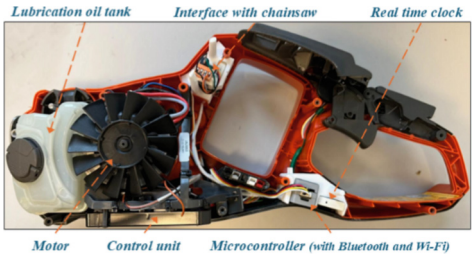



**Fig. 2.** Battery chainsaw sensing high-level architecture

This edge-preprocessing/cloud-arbitration split minimizes raw-data exposure while enabling accountable and auditable pricing as well as cumulative equity capping. Secure communication channels and authenticated endpoints protect telemetry in transit. Non-wear damage including cracks, oil leaks, and visible deformation is assessed upon tool return using a 360° computer-vision inspection pipeline, as illustrated for the battery chainsaw case in [21].

### 2.3 Framework for Fair Tool Sharing in Smart Neighborhoods

The framework targets common adoption barriers (trust deficits, late returns, and idle hoarding) by operationalizing an interpretable set of governance mechanisms. First, users are charged for possession time; charges are modulated by measured usage and estimated wear through a confidence-weighted blend of sensor-derived and task-based wear estimates. Second, holding a tool without productive use incurs stronger delay/idle penalties to discourage hoarding under contention. Third, past borrowing behavior (punctuality, care, and compliance) adjusts charges via a bounded multiplier. Fourth, the last compliant household in the queue receives a bounded discount to reduce stochastic inequities in waiting. GPS compliance is enforced via a tiered penalty, and all per-lease charges are bounded by spare-part and replacement caps and by a cumulative equity guardrail that limits household exposure. These components are formalized in Eqs., (detailed later). Governance workflow is illustrated in Fig. 3.

### 2.4 Related Work & Positioning

Prior work on IoT governance and blockchain accountability focuses on data provenance, device management, and secure interactions, but typically omits fairness metrics, queue

compensation, and sensor-informed wear integration for physical tools. The proposed framework introduces these missing dimensions by (i) applying Jain and Gini fairness measures to household service distribution (Jain's index summarizes how evenly the defined utilities are distributed across households, while the Gini coefficient summarizes dispersion/inequality of costs among served households), (ii) incorporating queue-position compensation, (iii) weighting wear charges by sensor-derived confidence, and (iv) imposing cumulative equity limits. Through these additions, the framework delivers a predictable and transparently fair pricing model designed to withstand both resource contention and adversarial user strategies (see Table 1).

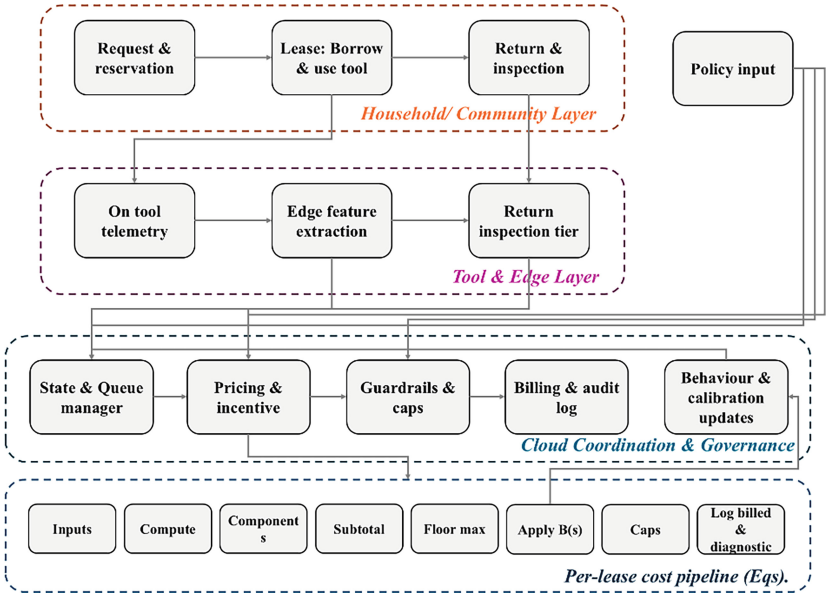



**Fig. 3.** End-to-end governance workflow (simplified): households request/lease tools; edge summaries feed the cloud service, which computes queue state, evaluates Eqs., applies caps, and logs billed/diagnostic records for audit and calibration

**Table 1.** Comparison of the proposed fairness-aware cyber-physical governance framework with existing IoT governance, blockchain accountability, and fair data-sharing approaches

| Category | Core focus | Limitations in Prior work | Proposed work contribution |
|---|---|---|---|
| IoT Governance Frameworks [22] | IoT standards coordination, device management, interoperability | Focuses on data and device governance, not physical resource circulation; no fairness metrics; no pricing or exposure caps | Governs physical tool circulation with time, wear, delay, behavior, GPS, and caps integrated into a unified cost model |

(*continued*)

**Table 1.** (*continued*)

| Category | Core focus | Limitations in Prior work | Proposed work contribution |
|---|---|---|---|
| Blockchain-Enabled IoT Governance [23] | Accountability and auditability via blockchain; secure interactions | Fairness not explicitly modeled; no queue compensation; no sensor-based wear integration; focuses on data provenance | Fairness-aware governance with explicit fairness metrics (Jain, Gini), queue compensation, and bounded exposure |
| Fair, Accountable Data Sharing in IoT [24] | Fair and secure data sharing among IoT clients; fee-based access to data | Operates at data layer, not physical tools; no wear, delay, or behavior modeling; fairness applies to data access, not resource circulation | Cyber-physical fairness: integrates sensing, behavior, delay pricing, and caps for physical resource allocation |
| IoT Resource Scheduling [25] | Scheduling idle IoT devices for computation tasks | Focuses on compute resources, not shared physical tools; no fairness or pricing; no behavioral incentives | Distributed scheduling with fairness for physical assets under contention, heterogeneity, and adversarial behavior |

### 2.5 Evaluation Design, Metrics, and Baselines

The evaluation combines analytical baselines with Monte Carlo simulation. Analytical baselines provide closed-form intuition for utilization and mean queueing time under Poisson arrivals and exponential service. The simulator implements the full cost model and evaluates four algorithmic baselines:

- FullModel: Complete pricing and allocation logic.
- TimeOnly: Pure time-based pricing.
- NoBehavior: Behavior multiplier removed.
- RandomAlloc: Capacity-feasible randomized allocation with full pricing.

For each configuration (supply $M$, adversarial fraction $f_{\text{adv}}$, behavior mode, confidence mode), 150 independent trials (fixed seed) are run, and mean $\pm$ 95% CI is reported for primary metrics: Jain's index on served-household utilities $u_i = 1/(1 + \text{cost}_i)$ (The study uses Jain's index on served-household utilities as the primary fairness metric because it is bounded in [0, 1] and scale-independent, capturing dispersion in realized experience among households that receive service. Because unserved households do not incur billed cost in the model, the study reports access impacts separately via fill rate and household coverage and treats all-household fairness as a secondary diagnostic) [26]. Gini coefficient computed on served-household costs (bounded in [0,1]). Clipped utilization, defined as busy time within horizon $H$ divided by $M \cdot H$; an extended horizon

is used as a sensitivity. Waiting time statistics (mean/median/95$^{\text{th}}$ percentile), measured relative to request time. Per-lease cost statistics (mean/std./95$^{\text{th}}$ percentile), reported as billed (capped) and diagnostic (uncapped) to separate policy truncation from mechanism effects. Cap-hit fraction, i.e., the fraction of leases hitting any cap, optionally broken down by spare-part, replacement, and per-lease equity cap hits. Request fill rate (fraction of requests served) and household coverage (fraction of households with $\geq 1$ lease).

The evaluation is a simulation-based design study intended to provide controlled comparative insights across capacities, adversarial fractions, and ablations. Because wear, behavior, and service processes are modeled rather than estimated from field data, the work emphasizes relative patterns (e.g., regime shifts and tail-truncation effects) rather than deployment performance claims.

Arrivals are generated during a 60-day window $H$. Scheduling and service may extend beyond $H$ to clear backlog; waits are measured relative to request times. Utilization is clipped by default; an extended horizon is used where noted. Queue length $q$ at request time is defined as the number of arrived requests that have not yet started service.

The last-compliant indicator $\chi \in \{0, 1\}$ is evaluated per lease using that lease's adversary flag and GPS status at the request time. The RandomAlloc baseline selects an available tool at arrival; if none are available, it selects among the earliest-free tools, ensuring capacity-feasible schedules and recomputed waiting times.

### 2.6 Queueing Intuition and Stability Regimes

To interpret observed delays across capacity settings, the simulator can be viewed through the lens of a multi-server FCFS queue with Poisson arrivals and approximately exponential service. With $N$ households and $M$ shared tools, arrivals per household follow a Poisson process with rate $\lambda_{\text{req}}$ requests/day; hence the aggregate arrival rate is $\lambda = N \cdot \lambda_{\text{req}}$ requests/day. Mean service time is $\mathbb{E}[T]$, giving a per-tool service rate $\mu = 1/\mathbb{E}[T]$ and total system capacity $M\mu$. The offered load is $\rho = \frac{\lambda}{M\mu}$. When $\rho \geq 1$, classical $M/M/c$ intuition suggests heavy queues and long waiting-time tails; when $\rho < 1$, delays remain bounded.

Under the settings used here (e.g., $N = 200$, $\lambda_{\text{req}} = 0.05$ requests/day, $\mathbb{E}[T] = 3$ days), one obtains $\lambda \approx 10$ requests/day and $\mu = 1/3$ day$^{-1}$, so $\rho = 30/M$. This yields representative regimes: $\rho \approx 6.0$ for $M = 5$ (severely overloaded), 3.0 for $M = 10$, 1.5 for $M = 20$, and 0.75 for $M = 40$ (stable).

Two utilization metrics are distinguished: clipped utilization and extended utilization, which accounts for backlog spillover beyond $H$. In overloaded regimes, extended measures expose saturation and long service spillovers even when clipped utilization appears near unity; in stable regimes, both metrics align and queues clear within the horizon.

## 3 Mathematical Framework

The coordination service computes an interpretable per-lease charge by aggregating time, usage, wear, inspection, delay, behavior, queue compensation, and GPS-compliance components, followed by the application of caps and a bounded behavior multiplier. Notation and default values are summarized in Table 2.

**Table 2.** Notation and default values (selected)

| Symbol | Meaning | Units/Range | Default / Notes |
|---|---|---|---|
| $N$ | Number of households | integer | 200 |
| $M$ | Number of shared tools | integer | 5, 10, 20, 40 |
| $H$ | Evaluation horizon | days | 60 |
| $T$ | Lease duration | days | Exponential, mean ≈3, min 0.5 |
| $T_0$ | Grace period | days | 2 |
| $\Delta$RUL | Measured wear | RUL units | Gaussian synthetic wear |
| $\Delta\text{RUL}_{\text{task}}$ | Task-based wear estimate | RUL units | $1.2 \times T$ |
| $c$ | Sensor confidence | [0, 1] | Beta($a$, $b$) |
| $s$ | Behavior score | [0, 1] | Beta($a$, $b$) |
| $x$ | Baseline fee per day | SEK/day | 12 |
| $\alpha$ | Wear → currency conversion | SEK/RUL | 2.0 |
| $V_{\text{spare}}$ | Spare-part cap | SEK | 200 |
| $V_{\text{tool}}$ | Tool replacement value | SEK | 6000 |
| $\lambda_{\text{eq}}$ | Equity cap fraction | [0, 1] | 0.5 |
| $\mathbb{E}[L]$ | Expected leases per household | – | $\lambda_{\text{req}}H(\approx 3)$ |
| Equity cap | Cumulative exposure | SEK | $\lambda_{\text{eq}}\mathbb{E}[L]V_{\text{spare}}$ |
| $v$ | GPS violation level | {0, 1, 2} | Random / adversarial |
| $\beta$, $\gamma$, $k$ | Delay kernel parameters | – | 2, 8, 0.25 |
| $L_{\text{week}}$ | Weekly idle cap | SEK | 150 |
| $\delta$ | Behavior sensitivity | – | 0.15 (clipped to [0.85, 1.15]) |
| $\phi_v$ | GPS penalty | SEK | {0, 50, 200} |
| Seed | RNG seed | integer | 20251217 |
| $f_{\text{adv}}$ | Adversarial fraction | [0, 0.2] | 0, 0.05, 0.10, 0.20 |

The time component measures how long a tool is held. A grace period $T_0$ allows short-term use without penalty. Excess holding time is defined as

$$D = \max(0, T - T_0) \tag{1}$$

where $D$ distinguishes timely returns from extended possession.

To distinguish active tool usage from idle possession, the framework applies a continuous usage weight. Let $\Delta$RUL denote the reduction in remaining useful life over the borrowing period. An activity threshold $\theta$ specifies the minimum average wear per day that qualifies as meaningful use. A sigmoid function with steepness parameter $a$ defines the usage weight:

$$w = \frac{1}{1 + \exp\left[-a\left(\frac{\Delta\text{RUL}}{\max(1, T - T_0)} - \theta\right)\right]} \tag{2}$$

For substantial wear, $w \to 1$, indicating active use; for minimal wear, $w \to 0$, indicating idle possession. Wear cost blends sensor-derived $\Delta$RUL with a task-based estimate $\Delta\text{RUL}_{\text{task}}$ using a confidence parameter $c \in [0, 1]$:

$$C_{\text{wear}} = \min\left(\alpha(c\ \Delta\text{RUL} + (1-c)\ \Delta\text{RUL}_{\text{task}}), V_{\text{spare}}\right) \tag{3}$$

Damage not captured by wear sensors (e.g., cosmetic damage or housing cracks) is handled via discrete inspection tiers $m \in \{0, 1, 2, 3\}$, each mapped to a surcharge $y_m$. The time-based cost component becomes

$$C_{\text{time}} = x\ T + y_m\ T \tag{4}$$

Delay pricing distinguishes between active and idle possession. Idle delay is penalized more strongly to discourage hoarding, especially when $M \ll N$. The idle penalty kernel is

$$C_{\text{idle}}(D) = \gamma\ e^{k\,\frac{e^{kD}-1}{e^{k}-1}} \tag{5}$$

where $k$ controls the growth rate. The total delay cost is

$$C_{\text{delay}} = \min\ (\beta D w + C_{\text{idle}}(D)(1-w), L_{\text{week}}) \tag{6}$$

Each user has a behavior score $s \in [0, 1]$. A bounded behavior multiplier adjusts charges:

$$B(s) = \text{clip}\ (1 - \delta(2s-1), B_{\min}, B_{\max}) \tag{7}$$

Queue fairness is addressed via compensation for the last compliant household in the queue. Let $q$ denote the queue length at request time, and let $\chi \in \{0, 1\}$ indicate whether the current lease was last among compliant participants:

$$Q(q, \chi) = \chi \cdot \eta \cdot x \cdot \min(1, \max(0, q-1)) \tag{8}$$

Equation (8) (queue compensation) is a bounded discount applied to the lease subtotal. The indicator $\chi$ activates compensation only for the household identified as the last compliant participant at request time; $\eta$ scales the discount relative to the baseline daily fee $x$, and $\min(1, \max(0, q-1))$ ensures the discount is zero when $q \leq 1$ and saturates for $q \geq 2$. The non-negativity floor in Eq. (11) prevents net credit payouts. For numerical safety in implementation, queue compensation is additionally capped so that $Q(q, \chi) \leq C_{time} + C_{wear} + C_{delay} + C_{gps}$, preventing compensation from exceeding the total of positive components before applying the non-negativity floor.

Geographic misuse is handled through a tiered penalty. Let $v$ denote the violation level and $\phi_v$ the associated penalty:

$$C_{\text{gps}} = \phi_v \tag{9}$$

The subtotal cost aggregates all components:

$$C_{\text{sub}} = C_{\text{time}} + C_{\text{wear}} + C_{\text{delay}} - Q(q, \chi) + C_{\text{gps}} \tag{10}$$

A non-negativity floor prevents net credit payouts:

$$C_{\text{sub}} \leftarrow \max(0, C_{\text{sub}}) \tag{11}$$

The final charge applies the behavior multiplier and caps, uncapped per-lease charge:

$$C_i^{unc} = \min\left(B(s_i)\ C_{sub,i}, V_{tool}\right) \tag{12}$$

Cumulative equity billing rule: let $Ecap = \lambda_{eq} E[L] V_{spare}$ and $S_i$ be cumulative billed cost so far:

$$C_i^{bill} = \min\left(C_i^{unc}, \max(0, Ecap - S_i)\right) \quad S_i \leftarrow S_i + C_i^{bill} \tag{13}$$

Expected leases per household over the horizon are

$$\mathbb{E}[L] = \lambda_{\text{req}} \cdot H \tag{14}$$

This parameterization, $\lambda_{\text{eq}} \mathbb{E}[L] V_{\text{spare}}$, aligns the equity cap with expected wear exposure and prevents population size $N$ from artificially suppressing household exposure.

## 4 Results and Discussion

Mean ± 95% CI over 150 trials per configuration for the baselines and metrics, focusing on fairness, queueing, utilization, and billed vs. diagnostic cost exposure across capacities $M \in \{5,10,20,40\}$ and adversarial fractions $f_{adv}$ is reported. Results are interpreted as comparative design insights under the stated simulation assumptions: regime shifts with capacity are robust within the modeled demand/service setting, while absolute levels and percentage reductions depend on the modeled wear, behavior, and penalty variability.

### 4.1 Overview of System Behavior

Across configurations, system behavior is primarily capacity-driven: low $M$ produces overload and long waits, while higher $M$ clears queues and improves service levels. Increasing capacity to $M = 40$ reduces clipped utilization to ≈0.72 at $f_{\text{adv}} = 0$, rising under adversarial activity (≈0.81 at moderate $f_{\text{adv}}$).

Served-household fairness at $M = 40$ is ≈0.255 for FullModel (versus ≈0.361 for TimeOnly). Queues are effectively cleared within the evaluation horizon (mean wait ≈0.01 days; p95 ≈0.07 days). Request fill rates remain ≈1.0 and household coverage ≈0.95 across baselines, indicating capacity-feasible scheduling and broad participation. Table 3 provides a summary comparison across baselines at $f_{adv} = 0$.

**Table 3.** Summary comparison across baselines at $f_{adv} = 0$ (key metrics; billed vs. diagnostic tails; cap-hit rate) (Values are trial means (150 Monte Carlo trials) for one representative behavior/confidence setting at $f_{adv} = 0$; 95% CI are computed over trials but omitted here for space).

| M | Baseline | Util (clipped) | Mean wait (d) | p95 wait (d) | Jain (served) | p95 cost billed (SEK) | p95 cost diagnostic (SEK) | Cap-hit frac (any) |
|---|---|---|---|---|---|---|---|---|
| 5 | FullModel | 0.995 | 150.67 | 289.32 | 0.055 | 273 | 396 | 0.134 |
| 5 | TimeOnly | 0.995 | 150.67 | 289.32 | 0.362 | 106 | 109 | 0.003 |
| 5 | NoBehavior | 0.995 | 150.67 | 289.32 | 0.054 | 286 | 422 | 0.159 |
| 5 | RandomAlloc | 0.995 | 150.67 | 289.32 | 0.055 | 273 | 396 | 0.134 |
| 10 | FullModel | 0.990 | 58.51 | 113.07 | 0.058 | 275 | 398 | 0.135 |
| 10 | TimeOnly | 0.990 | 58.51 | 113.07 | 0.359 | 106 | 107 | 0.003 |
| 10 | NoBehavior | 0.990 | 58.51 | 113.07 | 0.058 | 287 | 425 | 0.160 |
| 10 | RandomAlloc | 0.990 | 58.51 | 113.07 | 0.058 | 275 | 398 | 0.135 |
| 20 | FullModel | 0.976 | 14.39 | 28.24 | 0.047 | 275 | 398 | 0.135 |
| 20 | TimeOnly | 0.976 | 14.39 | 28.24 | 0.358 | 106 | 108 | 0.003 |
| 20 | NoBehavior | 0.976 | 14.39 | 28.24 | 0.046 | 287 | 422 | 0.159 |
| 20 | RandomAlloc | 0.976 | 14.39 | 28.24 | 0.048 | 274 | 398 | 0.135 |
| 40 | FullModel | 0.717 | 0.01 | 0.07 | 0.255 | 272 | 396 | 0.135 |
| 40 | TimeOnly | 0.717 | 0.01 | 0.07 | 0.361 | 106 | 108 | 0.003 |
| 40 | NoBehavior | 0.717 | 0.01 | 0.07 | 0.263 | 286 | 423 | 0.159 |
| 40 | RandomAlloc | 0.717 | 0.01 | 0.07 | 0.255 | 272 | 396 | 0.135 |

### 4.2 Fairness-utilization Trade-off

Figure 4 plots served-household Jain fairness against clipped utilization for all baselines and configurations. In the most constrained regimes (e.g., $M = 5$), FullModel clusters at high utilization but lower fairness, while TimeOnly often achieves higher fairness at similar utilization. This difference reflects the introduction of mechanism heterogeneity in FullModel (wear charges, inspection tiers, bounded behavior multipliers, and GPS penalties), which distributes costs more finely but can increase cost dispersion among served households.

As capacity increases to $M = 40$, FullModel fairness improves and converges toward TimeOnly; at $M = 20$ a dip is observed (Table 3), consistent with a transitional overloaded regime. RandomAlloc aligns closely with FullModel at each capacity, suggesting that pricing mechanics rather than modest allocation tie-breaks primarily drive fairness differences. Figure 5 isolates the $f_{\text{adv}} = 0$ slice and compares how fairness across all

households and the Gini dispersion of served-household costs evolve with capacity for FullModel and TimeOnly.

### 4.3 Queueing Performance versus Capacity

Figure 6 reports mean, median, and 95$^{\text{th}}$-percentile wait times as functions of capacity $M$ for FullModel with $f_{\text{adv}} = 0$. Waiting times decrease sharply with capacity. Because service is allowed to extend beyond the 60-day horizon, tails are substantial at low capacity ($M = 5$); at $M = 40$, the 95$^{\text{th}}$ percentile drops to $\approx$0.07 days.

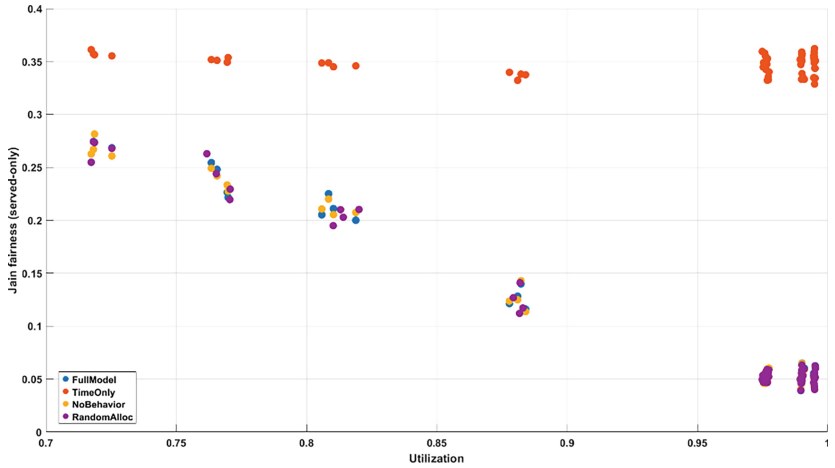



**Fig. 4.** Fairness versus utilization across baselines and configurations

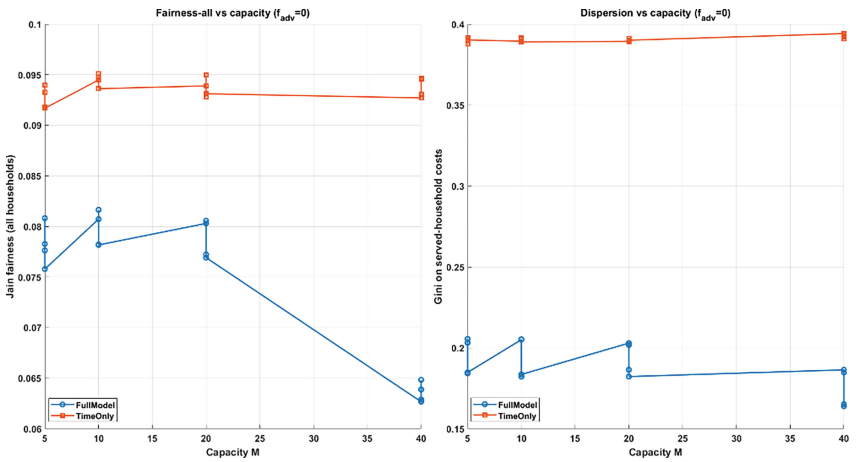



**Fig. 5.** Jain fairness (all households) and Gini dispersion as functions of capacity ($f_{\text{adv}} = 0$)

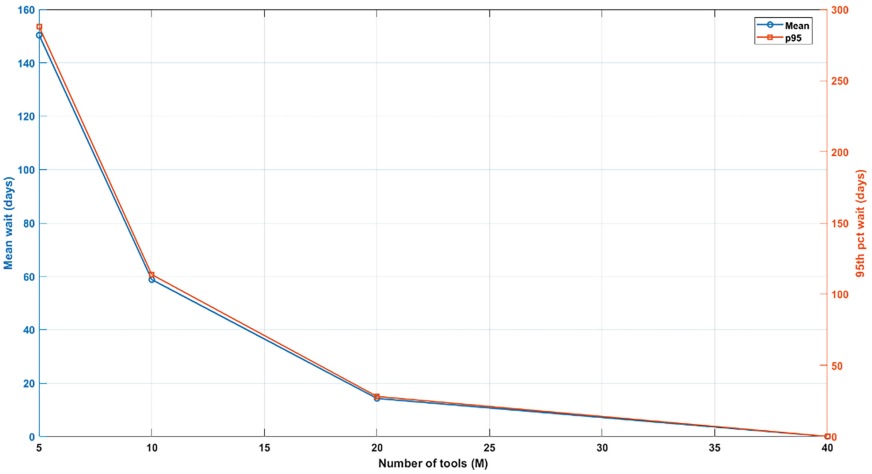



**Fig. 6.** Queueing outcomes versus capacity for all households (FullModel, $f_{\text{adv}} = 0$)

### 4.4 Policy versus Mechanism Effects Under the Equity Guardrail

Figure 7 contrasts diagnostic (uncapped) versus billed (capped) per-lease cost distributions under the cumulative equity cap $\lambda_{\text{eq}}\ \mathbb{E}[L]\ V_{\text{spare}}$. In stressed regimes (low $M$, higher $f_{\text{adv}}$), the guardrail truncates exposure tails, reducing billed 95$^{\text{th}}$-percentile costs by ≈35–50% relative to uncapped diagnostics (e.g., $M = 5, f_{\text{adv}} = 0.20$: billed p95 ≈297 SEK vs. uncapped p95 ≈523 SEK, a ≈43% reduction). Across all configurations, the reduction is ≈30–50%, while preserving mid-range sensitivity to usage, wear, and behavior.

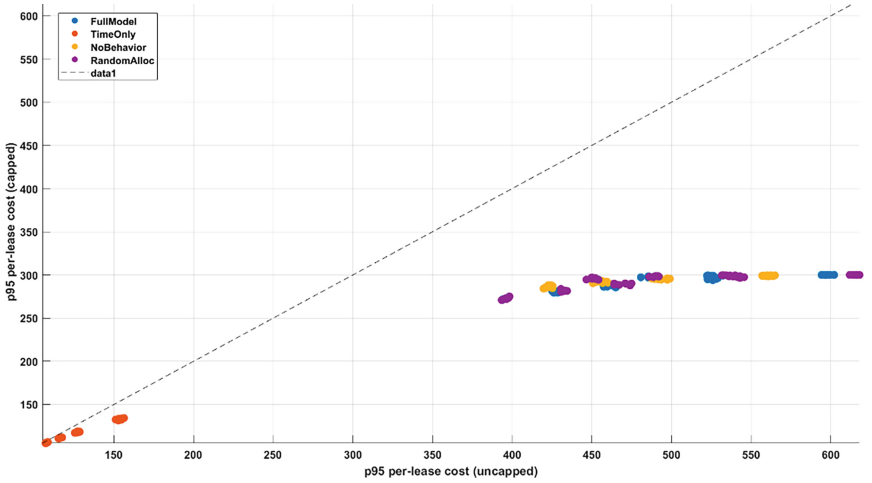



**Fig. 7.** Policy versus mechanism effects under the equity-cap guardrail

This truncation is reflected in the cap-hit fraction (any cap) for FullModel (≈0.13, roughly constant across capacities at $f_{adv} = 0$). TimeOnly exhibits near-zero any-cap hits (≈0.003).

### 4.5 Parameter Sensitivity and Calibration Strategy

The most influential parameters are $\alpha$ (wear-to-currency conversion), ($\theta$, $a$) (usage-weight sigmoid), and $\delta$ (behavior sensitivity) (See Table 4). Under tight capacity, reducing $\delta$ limits behavior-driven dispersion and can improve served-household fairness while maintaining deterrence for chronic misuse; increasing $\alpha$ strengthens wear accountability, while billed exposure remains bounded by the spare-part and equity caps. The pair ($\theta$, $a$) governs the idle-to-productive transition: larger $a$ sharpens the switch and larger $\theta$ shifts it toward higher wear/day.

**Table 4.** Practical calibration guide (illustrative)

| Parameter | Calibrate using (examples) | Minimal rule & sanity check |
|---|---|---|
| $\alpha$ | Spare-part prices; typical wear/lease; maintenance/inspection logs | Set $\alpha$ so typical wear maps to plausible SEK while respecting $V_{spare}$; check billed p95 and diagnostic tails behave sensibly. |
| $\theta$ | Wear/day proxies (telemetry); task labels or runtime/load proxies | Place $\theta$ at idle↔active boundary; check short productive tasks don't get $w \approx 0$. |
| $a$ | Spread/noise in wear/day proxies | Increase $a$ for sharper separation; decrease for smoother; check small noise doesn't cause large charge swings. |
| $\delta$ | Return punctuality; damage/compliance events; policy preference on dispersion | Start conservative under scarce capacity; increase if deterrence too weak; check served-household Jain doesn't collapse under tight capacity. |
| $\eta$ | Queue/wait distributions; policy preference for compensation | Choose $\eta$ so compensation is noticeable when queues form but bounded; check activation only when $q \geq 2$ and subtotal floor prevents negative bills. |

A practical workflow is: (i) choose a conservative $\delta$ to bound dispersion in scarce-capacity regimes, (ii) align $\alpha$ with observed wear/inspection outcomes (or maintenance proxies), (iii) tune ($\theta$, $a$) to reflect task profiles, and (iv) validate robustness using served-household fairness, billed tail metrics (e.g., p95), and cap-hit fractions (Table 3; Fig. 7).

## 5 Conclusion

This work introduced a governance framework for neighborhood tool sharing that combines time- and wear-aware pricing, behavior-based adjustments, queue-position compensation, GPS compliance, and multiple predictability caps. Applied to a battery-chainsaw case study and evaluated through Monte Carlo simulation, the framework

provides interpretable, bounded, and operationally feasible cost signals for distributed resource sharing.

Under tight capacity, mechanism-rich pricing can widen cost dispersion; this gap narrows as capacity increases. The cumulative equity guardrail effectively truncates extreme cost exposures, especially under adversarial behavior, while preserving meaningful mid-range signals tied to time, wear, and compliance.

Fairness is not strictly monotone in $M$: at $f_{adv} = 0$ the FullModel Jain index dips at $M = 20$ (Table 3), consistent with a transitional regime where the system remains overloaded ($\rho \approx 1.5$) but pricing heterogeneity still drives dispersion.

While increasing capacity $M$ reduces waiting times and can improve fairness in the modeled regimes, it also increases acquisition and maintenance costs borne by the operator. In practice, $M$ should be selected by trading off expected demand intensity (arrival rate) against a target service level (e.g., acceptable wait-time percentiles) under budget constraints. Seasonal demand (low- vs. high-season) or temporary demand spikes effectively increase the arrival rate and can push the system into an overloaded regime even when average demand appears stable; therefore, operators may combine moderate capacity with policies such as reservations, peak-period pricing, or temporary fleet augmentation to maintain predictability.

For municipalities, cooperatives, and OEM-supported sharing programs, the findings suggest a two-part strategy: (i) match capacity to demand intensity, and (ii) apply mechanism-rich pricing with transparent guardrails to deter misuse while ensuring predictable household exposure.

Overall, the study indicates that transparent, interpretable, and capped pricing can make tool sharing more trustworthy and scalable by linking charges to measurable usage, bounding downside risk, and providing operators with clear levers capacity, caps, and calibrated parameters to balance fairness, accountability, and efficiency.

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